

Adaptive Predictive Control-Based Noise Cancellation With Deep Learning for Arrhythmia Classification from ECG Signals

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Abstract—Arrhythmia is a disorder in the heart produced due to irregular electrical activities of the heart, and an electrocardiogram (ECG) represents a modality used by clinicians to discover arrhythmias. The occurrence of noise in ECG and irregular heartbeat is the complexity faced while diagnosing arrhythmias. Thus, there is a requirement for a technique that can attain high accuracy in arrhythmia classification. This paper utilizes the Deep Maxout network (DMN) for classifying arrhythmia using ECG signals. Here, the ECG signal is considered as an input to adaptive prediction noise control wherein the noise cancellation is performed using adaptive Least Mean Square (LMS) to discard noise and obtain a clean signal. From a clean signal, the features, like Empirical Mode Decomposition (EMD), wave component detection, wave features, such as RR interval, QT interval, PR interval, PP interval, and R peak whereas statistical features, such as kurtosis, eccentricity are mined for improved analysis. At last, the classification of arrhythmia is done using DMN. The greatest accuracy of 92.1%, the sensitivity of 92.6%, and the specificity of 91.9% are measured using DMN.

Index Terms—Adaptive predictive noise control, Arrhythmia classification, Deep Maxout network, Electrocardiogram signal, Wave component detection

I. INTRODUCTION

ECG is an imperative application, which recorded the procedure of transmission, cardiac excitability, and recovery. The automated discovery of uneven rhythms of the heart via ECG is a major chore in automated treatment of cardiac disorders [1]. ECG is a general and non-invasive treatment model for detecting heart issues, like arrhythmias [2] [3]. The arrhythmia in heart disease causes some trouble in the usual rate of the heart wherein the heart can beat too gradually or too quickly or erratically. The arrhythmias can lead to the indication that involves palpitations, and breath conciseness. Several arrhythmias are not severe, but some arrhythmias can lead to heart failure, cardiac arrest, and stroke [4] [5]. ECG signal is an imperative part of observing disorders of the heart and arrhythmia discovery as they trace all cardiac tasks and involve data regarding the functioning of the heart [6].

Assessment of long-time ECG data is a slow and laborious task. Therefore, automatic discovery and analysis of various kinds of arrhythmias using slight variations in duration, morphology, and ECG amplitude are of huge support to clinicians [7]. QRS/PT wave detection and parallel Delta modulation are used in real-time machine learning for arrhythmia classification in [8]. A front-end IC, an FPGA-based delineation algorithm, and an arrhythmia classification algorithm are all components of an ECG delineation and arrhythmia classification system [9]. The machine learning techniques are adapted for detecting arrhythmia that involves Decision Tree (DT) [10], Artificial Neural Networks (ANN), Random Forest (RF) [11], K-Nearest Neighbor (KNN) [12], Ensemble learning [13], Support Vector Machine (SVM), Adaptive Neuro-Fuzzy Inference System (ANFIS), and Deep learning [14] [15]. The dynamic fuzzy logic module described in [16] [17] is based on two fuzzy approximators and is designed to change how input-output fuzzy membership functions are supported in the inference engine module. Arrhythmia can be prevented and treated early with intelligent ECG analysis [18].

In [19], deep two-dimensional CNN was utilized with grayscale ECG for classifying arrhythmia. The accuracy of classification was enhanced with major steps, like augmentation of data to enlarge training data [20] [21]. It is suggested to classify arrhythmias using deep neural networks (DNNs). To extract characteristics from unprocessed ECG signals, two DNN models are trained using bidirectional LSTM layers and residual convolutional modules. For classification, the retrieved features are combined into a feature vector [22]. There exist various techniques adapted for automated classification of ECG such as frequency assessment, Support Vector Neural Network (SVNN), CNN, mixture-of-experts techniques, Probabilistic Neural Networks, KNN clustering, ANN and Hidden Markov Models (HMM), Classification and Regression Trees, Recurrent Neural Networks (RNN).

The aim is to design a model to classify arrhythmia with ECG signals. The main contribution is the design of an

arrhythmia classification model using a DMN and adaptive predictive control-based noise cancellation model. Here, the input ECG signal is obtained from a database and adaptive predictive control-based noise cancellation is done using adaptive Least Mean Square (LMS). The noise-free output signal is obtained from the above process. Then, the feature extraction is carried out whereas unique features including EMD, statistical features, and wave features are extracted for the further classification process. Finally, arrhythmia classification is done using DMN with the extracted features.

The major contribution of the paper is

DMN for classifying arrhythmia with ECG: The DMN is adapted for classifying arrhythmia with ECG. Here, the set of statistical, wave, and EMD features are mined from ECG for classifying arrhythmia.

The remaining sections are enlisted. Section 2 describes the assessment of arrhythmia classification techniques. Section 3 presents DMN for performing arrhythmia classification. Section 4 describes the efficiency of DMN in relating with other classifiers. The conclusion is presented in Section 5.

II. MOTIVATIONS

This section examines classical arrhythmia classification techniques along with their challenges. The existing arrhythmia classification techniques are designed based on limited training data, which lacks adaption of particular features of the rhythm of patients which can be varying according to training data. Thus, the present techniques cannot execute effectively in real conditions. The above-mentioned issue and problems of existing techniques are termed as motivation for designing a new model.

A. Literature Survey

The illustration of eight classical arrhythmia classification methods with ECG signal is revealed along with its advantages and disadvantages. Jingshan Huang et al. [1] advanced a system based on CNN for arrhythmia classification using ECG signal. Here, a short-time Fourier transform was used to transform the time domain signals into time-frequency spectrograms. At last, 2D-CNN was used to classify arrhythmias. However, this technique affected the performance of the system due to the fluctuation of various learning rates. Ehab Essa et al. [3] devised a deep learning-based multi-model system for classifying arrhythmia using ECG. Here, two various deep learning bagging models were devised for classifying heartbeats into various kinds of arrhythmias. The first model united long short-term memory (LSTM) and CNN for capturing temporal dynamics and local features. The second model RRHOS-LSTM combined conventional features, like higher-order statistics (HOS) and RR intervals, with the LSTM model for classifying arrhythmias.

However, the method did not localize feature points. Runnan He et al. [22] utilized bidirectional long short-term memory (BiLSTM) and Deep Neural Networks (DNN) for arrhythmia classification using ECG signal. BiLSTM layers were skilled in mining features from raw ECG signals. Obtained features

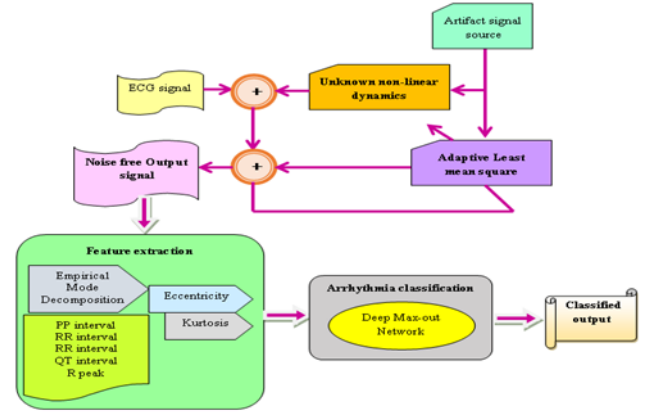


Fig. 1. Structure of Arrhythmia classification model using DMN

were combined for establishing a feature vector to classify arrhythmias. However, the irrelevant data affected the system's performance. Wei Zeng and Chengzhi Yuan [15] devised a technique, namely variational mode decomposition (VMD) for determining cardiac arrhythmia using an ECG signal. Here, the VMD was adapted for decomposing the ECG into various intrinsic modes. The neural networks were utilized for classifying the arrhythmia using ECG. The technique poses a limited database size.

B. Challenges

The issues confronted by classical arrhythmia classification methods using ECG signals are as follows,

- The amount of data adapted for analyzing the ECG is high. Thus, there exists an elevated probability to have a false analysis of arrhythmia. Moreover, the indication of arrhythmia cannot be seen throughout the ECG signal capturing period [23].
- In [15], VMD is devised for classifying arrhythmia with ECG signal. However, this technique did not involve other advanced features, like the Hurst exponent, and recurrence quantification for improving the performance of classification.
- The properties of ECG signal, like amplitude and period, vary amongst the person with various aspects, like gender, age, and physical conditions. Hence, determining a comprehensive platform along with the associated standards is a challenging task [23].

III. DEEP MAXOUT NETWORK FOR CLASSIFYING ARRHYTHMIA WITH ECG

Arrhythmia represents cardiovascular disease caused due to heart problems. The automated discovery of uneven heart rhythms using ECG signals is a major task for the automated treatment of cardiovascular disease. The aim is to develop a method for arrhythmia classification with ECG signals. The goal is to devise a deep network for classifying arrhythmia. Here, the input ECG signal is considered as an input to adaptive predictive noise control [24]. In adaptive predictive

noise control, the noise cancellation is done using adaptive LMS wherein the noise-free output signal is obtained. Then, the extraction of features is done in which certain features, like EMD, wave features, and statistical features are mined for improved arrhythmia classification. At last, the classification of arrhythmia is done with DMN considering obtained features. The architecture of the arrhythmia classification model using DMN is revealed in Fig. 1.

Assume a dataset, which comprises ECG signals expressed as that consists of ECG signal, that is expressed as,

$$E = \{G_1, G_2, \dots, G_p, \dots, G_q\} \quad (1)$$

here, q refers to total ECG signals in dataset, and G_p signifies p^{th} signal.

A. Adaptive Predictive Control-based ECG Noise Cancellation

The inputted ECG signal G_p is considered an input to adaptive prediction noise control [24]. Here, the noise cancellation is applied to the ECG signal for discarding noise to make the signal appropriate for further processing. Adaptive noise cancellation is used for discarding noise from the ECG signal. The process of adaptive noise cancellation needs two inputs. The first input is produced with an ECG signal and expressed as $M(o)$. The second input is accumulated from the artifact signal and expressed as $N(o)$. Noise source adapted signifies the origin in which several artifacts, like EOG, ECG, and EMG produce. Noisy source signal representing artifacts signal is fed to anonymous non-linear dynamics that resulted in interference signal $P(o)$. Thereafter, the integration of both clean signal and interference signal produces the primary input signal, expressed as,

$$Q(o) = M(o) + P(o) \quad (2)$$

Where, $M(o)$ refers to input source signal and $P(o)$ express interference signal, which is produced from noise. On the other hand, the signal produced with the noise source undergoes an adaptive filtering process for attaining adaptive LMS filter output. The output of the adaptive LMS filter is nearer to the interference signal, which is produced with nonlinear dynamics. The goal of adaptive noise cancellation is done for obtaining a clean ECG. For retrieving a clean ECG signal, the output of the adaptive LMS filter undergoes subtraction with primary input, which is expressed as,

$$M_r(o) = M(o) + P(o) - H(o) \quad (3)$$

Where, $H(o)$ symbolize output of adaptive LMS filter, and $M_r(o)$ signifies output signal of complete noise cancellation. Thus, the obtained noise-free output signal is expressed as F . An Adaptive LMS filter provides the feasibility to discover the changes of stiffness with nonlinear structures with respect to time. The adaptive LMS filter tends to be computationally effective and functions in a robust manner. An adaptive LMS filter represents a digital filter wherein the coefficients can alter with respect to time to enhance its performance, thereby allowing the filter to employ the changes in the noisy input

signal. The filter coefficients are modeled on basis of the sample to reduce the error signal between the measured and the modeled output signal.

B. Extraction of Significant Features

In the feature extraction process, the noise-free ECG signal F obtained from adaptive prediction noise control is considered as input where features are mined from each ECG. Thus, imperative features are generated using an ECG signal, and mining insinuation is inspected. Feature mining is done for yielding pertinent features to deliver enhanced arrhythmia classification. Features mined with adaptive prediction noise control involve wave component detection, statistical features, and wave features. Every feature mined by ECG signal is defined under:

1) *Wave Component Detection* : Wavelet component detection [25] technique undergoes multi-resolution wavelet-based technique where Eight wavelet levels $D[K^l]$ are found in the eight-level decomposition, which is based on the l^{th} patient's ECG data. The breakdown of wavelet in ECG produces eight scales of detail coefficients. The small scale signifies high-frequency components, and the high scale signifies the signal's low-frequency components. Q-wave, R-wave, and S-wave (QRS) contain fewer frequencies. Thus, high decomposition degree utilized low-frequency units of ECG. The outcome of decomposition is given as,

$$|a_1^l, a_2^l, \dots, a_8^l = D[K^l]| \quad (4)$$

where a_1^l, a_2^l, a_8^l signifies coefficient of reconstruction wavelets with respect to ECG of l^{th} patient. The process of determining the ECG waves is as follows: At first, the R peak is determined using the suitable choice of wavelet coefficients followed by detecting S and Q points with five-point differentiation. At last, wavelet coefficients of S and T points are discovered.

a) *R-peak*: The wave is reconstructed for discovering QRS is expressed as,

$$m_1^l = a_3^l + a_4^l + a_5^l \quad (5)$$

where a_3^l, a_4^l, a_5^l signifies reconstruction coefficients of l^{th} patient. For discovering the R peak, the waves reconstructed are utilized as there exist oscillations in QRS, which is expressed as,

$$m_2^l = \frac{a_4^l \times (a_3^l + a_5^l)}{2^s} \quad (6)$$

Where signifies decomposition level and its value is eight. The modulus $|m_1^l \times m_2^l|$ is discovered, and the wave with the highest amplitude signifies the R peak.

b) *S and Q Waves*: S and Q waves contain elevated frequency and less amplitude using their energy focused on small-scale wavelets, and thus reconstruction coefficients are adapted for reconstructing wave, which is expressed as,

$$m_3^l = a_2^l + a_3^l + a_4^l + a_5^l \quad (7)$$

c) *T and P Waves*: The waves reconstructed for discovering T and P waves are given as,

$$m_4^k = a_6^k + a_7^k \quad (8)$$

The T wave is discovered as the highest point once the S point is determined and crossing points having the highest ability are discovered as T-onset and T-offset points. The output of wavelet component detection is expressed as J .

2) *Wave Features*: The PP interval, PR interval, R-peak interval, QT interval, and RR interval are among the wave characteristics that are mined. Below is a summary of each feature.

a) *PP Interval*: PP interval is a time instance amongst dual successive P waves contained in ECG. Consider count of P points alters amongst l and v as expressed as,

the PP interval represents the duration of time amongst dual consecutive P waves contained in ECG. Take count of P points alters amongst l and v is expressed as

$$PP = \{PP_h \quad ; 1 \leq h \leq g\} \quad (9)$$

$$PP = \frac{1}{g} \sum_{h=1}^g [P_l^h - P_l^{h+1}] \quad (10)$$

Where, P_l^h signifies PP component linked with h , P points of l^{th} patient. P_l^{h+1} symbolize PP component linked to $(h+1)^{th}$ point of l^{th} patient. PP interval of g points is averaged to establish the PP feature.

b) *PR interval*: PR interval expresses interval of time amongst P wave of h^{th} point and R wave of $(h+1)^{th}$ point. Moreover, the PR interval represents the duration of time between the P wave of the point and the R wave of the consecutive point. The average of all the points is found for exhibiting the PR feature of the ECG signal.

$$PR = \{PR_h \quad ; 1 \leq h \leq g\} \quad (11)$$

$$PR = \frac{1}{g} \sum_{h=1}^g [P_l^h - R_l^{h+1}] \quad (12)$$

where P_l^h symbolize P Wave of h^{th} point of the l^{th} patient and R_l^{h+1} express R wave of $(h+1)^{th}$ point of l^{th} patient.

c) *RR Interval*: RR interval represents an instance of time amongst consecutive R waves contained in the ECG of the l^{th} patient. RR interval of all g points are discovered and averaged for establishing the RR feature of ECG given as,

$$RR = \{RR_h \quad ; 1 \leq h \leq g\} \quad (13)$$

$$RR = \frac{1}{g} [R_l^h - R_l^{h+1}] \quad (14)$$

where R_l^h symbolize R Wave of h^{th} point of the l^{th} patient and R_l^{h+1} express R wave of $(h+1)^{th}$ point of l^{th} the patient.

d) *QT Interval*: QT interval signifies time instance amongst Q wave and T wave of ECG of l^{th} patient. Here, all QT intervals contained in signals of ECG of l^{th} patient are averaged for representing the QT feature of ECG given as,

$$QT = \{QT_h \quad 1 \leq h \leq g\} \quad (15)$$

$$QT = \frac{1}{g} \sum_{h=1}^g [Q_l^h - T_l^{h+1}] \quad (16)$$

where Q_l^h symbolize Q Wave of h^{th} point of l^{th} patient and Q_l^{h+1} express Q wave of $(h+1)^{th}$ point of l^{th} patient.

e) *R-Peak*: It refers extreme amplitude of the R wave averaged for establishing ECG's R peak feature

R peak denotes the highest amplitude value in R wave averaged for establishing R peak of ECG given as,

$$R = \frac{1}{g} \sum_{h=1}^g R^h \quad (17)$$

Hence, the wave consists of time instance features linked to g points contained in the ECG signal of the l^{th} patient expressed below,

$$w = \{PP^l, PR^l, RR^l, QT^l, R^l\} \quad (18)$$

here, PP^l is the PP interval feature of the l^{th} patient, PR^l is PR interval feature of the l^{th} patient, RR^l is RR interval feature of the l^{th} patient, QT^l is QT interval feature of the l^{th} patient, and R^l is R peak feature of the l^{th} patient. Here, the wave feature is expressed as ω .

3) *Statistical Features*: Statistical features are used for manipulating data by evaluating and drawing conclusions from data. The statistical features involve kurtosis, and eccentricity, which are discussed below.

a) *Eccentricity*: Eccentricity is the measure of the difference between the foci ellipses of an item and a small foci ellipse in terms of distance. Eccentricity has a value of between 0 and 1. Straight objects show eccentricity as 1, whereas those that are closer to the circle show eccentricity as 0. This eccentricity feature is expressed as ϵ .

b) *Kurtosis*: The kurtosis refers to a statistical measure, which describes the extreme values that are the peak of frequency distribution. This kurtosis feature is expressed as κ .

The statistical feature is expressed as S , that is formulated as,

$$S = \{\epsilon, \kappa\} \quad (19)$$

4) *Empirical Mode Decomposition*: To locally divide superposed spatial frequencies in progress images, EMD [26] is performed in the Hilbert-Huang transform (HHT). Signals in the temporal domain are evaluated using it. The purpose of EMD is to build a surface for 2D scattered data points that refer to an extreme point. Each data should include surfaces that are interposed. Intrinsic Mode Functions (IMF), which are formed during the signal and sifting process, are frequency

units with the property of splitting the signal into them. The sifting method splits the sampled signal using dual restrictions and EMD. Each IMF in this instance has a comparable number of crossing zeros and extrema. IMF recognizes that there are at least two extremes and balances based on the local mean. The phrase "EMD features" refers to features produced by the filtering process.

5) *Configuration of the Feature Vector*: Hence, the feature vector contains the wave features, statistical features, and wavelet component detection feature, which is expressed as,

$$A = \{S, \omega, J, B\} \quad (20)$$

where, refers to statistical feature, ω express wave feature, J wavelet component detection feature, and refers EMD feature.

6) *Arrhythmia Classification using DMN*: The classification of arrhythmias is done using DMN considering the obtained feature vector denoted as A . The structure of DMN is exhibited in Fig 2. Multi-layer structured Maxout with trainable activation function approximators and non-saturated activation functions are what is referred to as DMN. DMN offers the procedure of optimization by averting hidden units. Despite, the Maxout unit signifies a trainable activation role, it's not an arbitrary function approximator. Hence, input provided to DMN has extracted features A . The DMN is effective in approximating random activation functions. Fig. 2 signifies the structure of DMN.

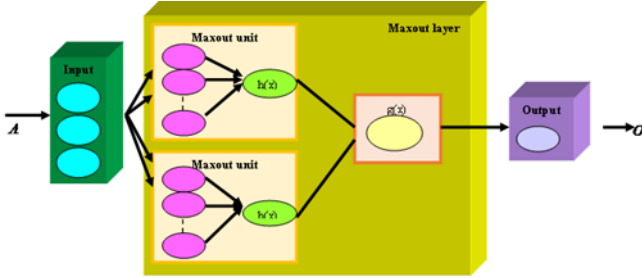


Fig. 2. Structure of DMN

Assume features as input $Z \in R^d$ where Z symbolize raw input vector, and activation of the hidden unit is given by,

$$d_{i,j}^m = \max_{j \in [1, k_m]} d_{i,j}^{m-1T} \varpi_{i,j} + e_{i,j} \quad (21)$$

$$h_i = \max_{j \in [1, k_n]} d_{i,j}^n \quad (22)$$

Where n signifies total count of layers, $\varpi_{i,j}$ express weights, and e refers to a bias in DMN. The output of DMN is given by O , that can classify arrhythmia.

IV. RESULTS AND DISCUSSION

Efficiency of DMN is assessed by considering accuracy, sensitivity, and specificity by changing training data and K-value.

A. Experimental Setup

Developed DMN is applied on Windows 10 OS that has an Intel core processor and also 2GB RAM and is performed in Matlab.

B. Dataset used

Dataset utilized for assessment and categorization is The MIT-BIH Arrhythmia Database and Normal Sinus Rhythm Database, which is combined for performing the assessment by extracting features. MIT-BIH Normal Sinus Rhythm Database [27] comprises 18 recordings of ECG and people generated from the Arrhythmia Laboratory at Boston's Beth Israel Hospital. People adapted in dataset 5 males and 13 women, all between the ages of 20 and 50, are involved. The Boston Beth Israel Hospital's MIT-BIH Arrhythmia Database [28], which includes 48 half-hour snippets of two-channel ambulatory ECG recordings, is taken into consideration.

C. Investigational Outcomes

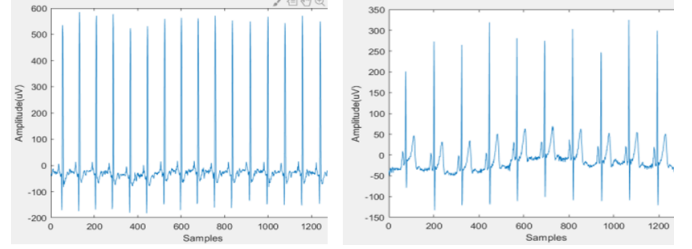


Fig. 3. Investigational outcomes of DMN by Input normal signal

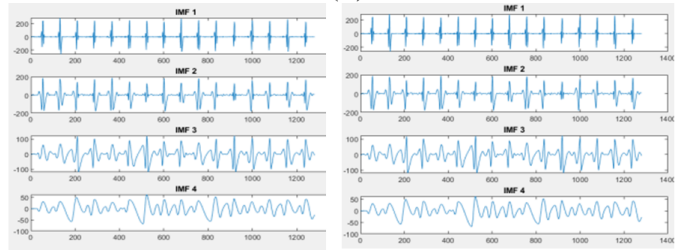


Fig. 4. Investigational outcomes of DMN by EMD feature for normal signal

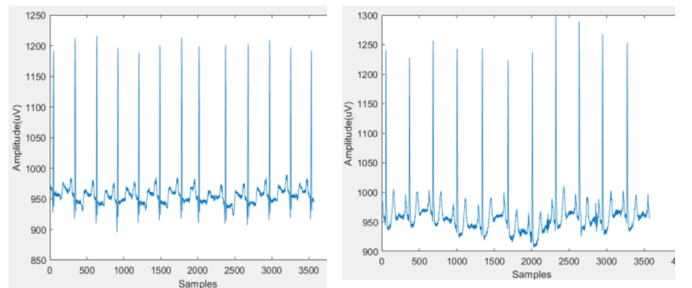


Fig. 5. Investigational outcomes of DMN by input Arrhythmia signal

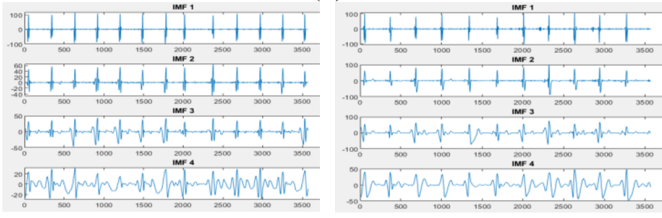


Fig. 6. Investigational outcomes of DMN by EMD feature for Arrhythmia signal

Fig.3, Fig.4., Fig.5., Fig.6. Reveals experimental results of DMN using ECG signal for classifying arrhythmia. The input ECG normal signal is exhibited in Fig. 3. EMD feature for inputted normal ECG is exposed in Fig. 4. Input arrhythmia ECG is displayed in Fig. 5. The EMD feature for inputted arrhythmia ECG signal is depicted in Fig. 6.

D. Evaluation Metrics

Adaption of developed arrhythmia categorization is done by certain measures, that are defined below.

1) *Accuracy*: This indicates the degree of the nearness of value that is computed with respect to the original value in arrhythmia categorization, and is expressed as,

$$Accuracy = \frac{Y^P + Z^n}{Y^P + Y^n + Z^P + Z^n} \quad (23)$$

Here, Y^P means true positive, Y^n denotes true negative, Z^P mentions false positive, and Z^n means false negative.

2) *Sensitivity*: It designates the ratio of positives detected using the arrhythmia categorization technique appropriately, and is expressed as,

$$Sensitivity = \frac{Y^P}{Y^P + Z^n} \quad (24)$$

3) *Specificity*: This indicates the negative ratio detected by the proposed model exactly and expressed as,

$$Specificity = \frac{Y^n}{Y^n + Z^P} \quad (25)$$

E. Comparative Approaches

Techniques taken for assessment comprises SIFT+CNN [1] DRN+LSTM [22], CNN+LSTM [29], and DMN.

F. Comparative Analysis

Evaluation of methods by sensitivity, accuracy, and specificity is defined by shifting K-fold and training data.

1) *Evaluation with Training Data* : Technique assessment by changing training data is displayed in Fig. 7., Fig. 8, Fig. 9. Evaluation by Fig.7. Illustrates accuracy. Accuracy for 50% of the training data measured using SIFT+CNN, DRN+LSTM, CNN+LSTM, and DMN. 0.719, 0.720, 0.721, and 0.745, respectively. Additionally, the accuracy scores obtained by SIFT+CNN, DRN+LSTM, CNN+LSTM, and DMN with 90% training data are 0.855, 0.872, 0.887, and 0.921. The enhancement of SIFT+CNN, DRN+LSTM, and CNN+LSTM performance, in accordance with DMN using accuracy, are

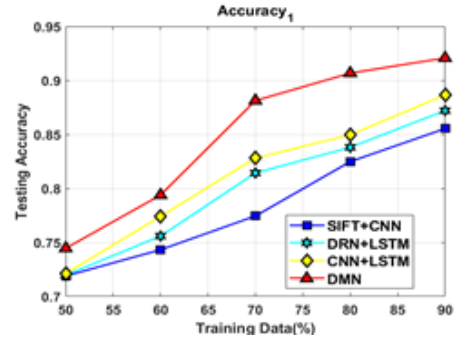


Fig. 7. Assessment of methods with training data by Accuracy

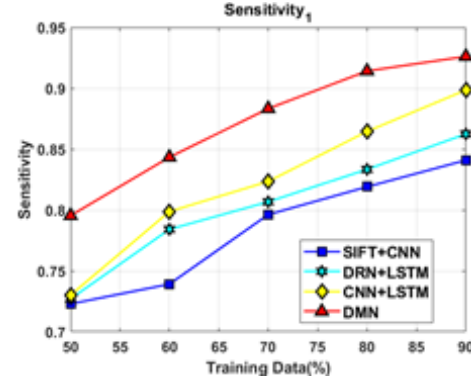


Fig. 8. Assessment of methods with training data by Sensitivity

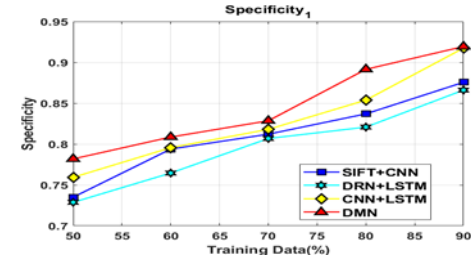


Fig. 9. Assessment of methods with training data by Specificity

7.166%, 5.320%, and 3.691%. Fig.8. Shows the assessment using sensitivity. Sensitivity values calculated with SIFT+CNN, DRN+LSTM, CNN+LSTM, and DMN are 0.723, 0.727, 0.730, and 0.795 for 50% training data, respectively. Additionally, the sensitivity values calculated by SIFT+CNN, DRN+LSTM, CNN+LSTM, and DMN for 90% training data are 0.841, 0.862, 0.899, and 0.926, respectively. Performance enhancement of SIFT+CNN, DRN+LSTM, and CNN+LSTM, concerning DMN using sensitivity is 9.179%, 6.911%, 2.915%. Fig.9. Illustrates evaluation by specificity. Specificity values for SIFT+CNN, DRN+LSTM, CNN+LSTM, and DMN for 50% training data are 0.735, 0.729, 0.759, and 0.782, respectively. Additionally, the specificity qualified by SIFT+CNN, DRN+LSTM, CNN+LSTM, and DMN are 0.876, 0.866, 0.917, and 0.919 for 90% training data, respectively. Performance enhancement of SIFT+CNN, DRN+LSTM, and

CNN+LSTM, concerning DMN using specificity are 4.678%, 5.767%, and 0.217%.

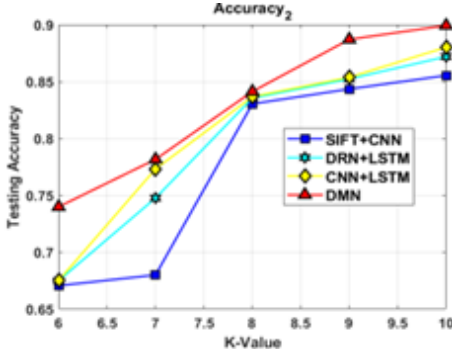


Fig. 10. Assessment of methods with K-value by Accuracy

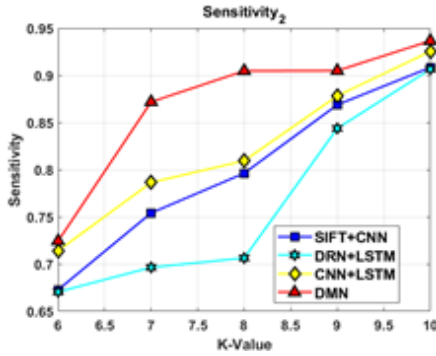


Fig. 11. Assessment of methods with K-value by Sensitivity

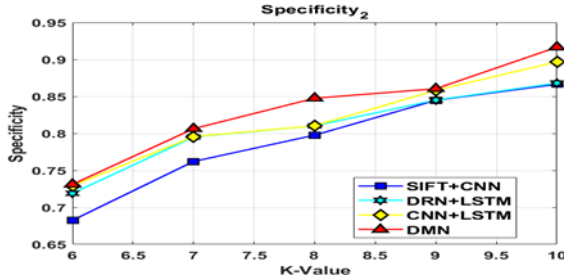


Fig. 12. Assessment of methods with K-value by Specificity

2) *Assessment with K-fold* : Fig.10, Fig.11, Fig.12 reveals the evaluation of methods by changing K-value. Assessment by accuracy is showed in Fig.10. For K-value=6, accuracy assessed using SIFT+CNN, DRN+LSTM, CNN+LSTM are 0.671, 0.675, 0.675, while that of DMN is 0.740. Also, for K-value=10, the accuracy measured by SIFT+CNN, DRN+LSTM, CNN+LSTM are 0.855, 0.872, 0.880, while that of DMN is 0.899. Performance enhancement of SIFT+CNN, DRN+LSTM, and CNN+LSTM, in accordance to DMN by accuracy are 4.894%, 3.003%, 2.113%. Fig.11 shows the assessment with sensitivity. Sensitivity measurements utilising SIFT+CNN, DRN+LSTM, CNN+LSTM, and K-value=6 are 0.673, 0.670, and 0.714, respectively, whereas DMN is

0.724. Moreover, for K-value=10, sensitivity measured using SIFT+CNN, DRN+LSTM, CNN+LSTM are 0.908, 0.907, 0.925, while that of DMN is 0.937. The performance improvement of SIFT+CNN, DRN+LSTM, and CNN+LSTM, with respect to DMN using sensitivity are 3.094%, 3.201%, and 1.280%. Evaluation by specificity is exposed in Fig.12. For K-value=6, specificity measured using SIFT+CNN, DRN+LSTM, CNN+LSTM, are 0.683, 0.719, 0.729, while that of DMN is 0.731. Moreover, for K-value=10, the specificity measured using SIFT+CNN, DRN+LSTM, CNN+LSTM are 0.866, 0.868, 0.897, whereas DMN's is 0.917. With regard to DMN employing specificity, SIFT+CNN, DRN+LSTM, and CNN+LSTM have improved performance by 5.561%, 5.343%, and 2.181% respectively.

G. Comparative Discussion

Table I describe the comparative assessment of techniques for altering k-fold and training data. For training data, maximum accuracy of 0.921 is evaluated using DMN while that of SIFT+CNN, DRN+LSTM, and CNN+LSTM are 0.855, 0.872, and 0.887. The maximum sensitivity of 0.926 is measured by DMN while that of SIFT+CNN, DRN+LSTM, and CNN+LSTM are 0.841, 0.862, and 0.899. The highest specificity of 0.926 is measured by DMN while that of SIFT+CNN, DRN+LSTM, and CNN+LSTM are 0.876, 0.866, and 0.917. For K-fold, maximum accuracy of 0.899, maximum sensitivity of 0.937, and maximum specificity of 0.917 are evaluated using DMN.

TABLE I
ASSESSMENT COMPARISON

Variation	Metrics	SIFT+ CNN	DRN+ LSTM	CNN+ LSTM	DMN
Training data	Specificity	0.876	0.866	0.917	0.919
	Sensitivity	0.841	0.862	0.899	0.926
	Accuracy	0.855	0.872	0.887	0.921
K-fold	Specificity	0.866	0.868	0.897	0.917
	Sensitivity	0.908	0.907	0.925	0.937
	Accuracy	0.855	0.872	0.880	0.899

V. CONCLUSION

The assessment of huge-time ECG is a labourious and complex task. Hence, automated discovery and analysis of several kinds of arrhythmias based on slight variations in duration, morphology, and ECG amplitude are of great support to medical practitioners. A DMN is utilized to classify the arrhythmia with an ECG signal. Here, the ECG signal is taken as input to adaptive prediction noise control in which the cancellation of noise is carried out using adaptive LMS to discard noise and generate a clean signal. Using clean signal, the features such as EMD, wave component detection, statistical features like kurtosis, eccentricity, and wave features like PR, PP, RR, and QT intervals along with the R peak are mined for making effective arrhythmia classification. At last, the categorization of arrhythmia is done with DMN. The

DMN is easily understandable by various kinds of people and it is effective in making effective decisions. The maximum accuracy of 92.1%, the sensitivity of 92.6%, and the specificity of 91.9% are measured by DMN. In the future, feasibility will be checked with other dataset for proposed model.

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